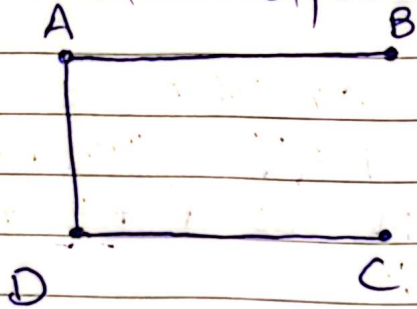


Date

10.05.19

Graphs: A graph = $G(V, E)$ consists of V , a non empty set of vertices and E a set of Edges. Edges connect either one or two vertices, called endpoints-



- Simple graph :
 - each edge connects two different vertices.
 - no two edges connect the same two vertices

Multigraph:

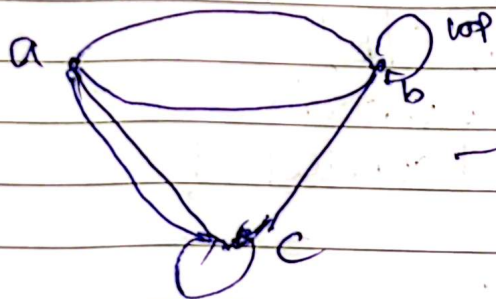
- multiple edges connecting same two vertices

Loop:

- An edge that connects a vertex to itself.

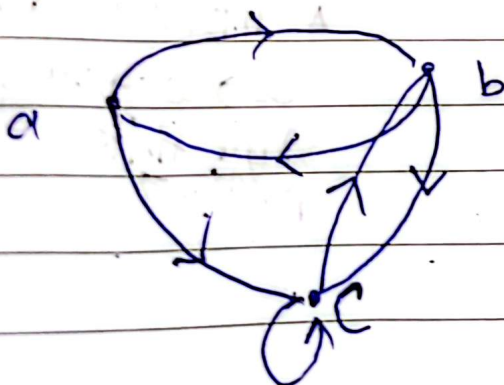
Pseudograph:

- may include loops as well as multiple edges



Directed graph - Digraph

graph $G = (V, E)$ where each edge is associated with a pair of vertices. The directed edge (a, b) begins at a and ends at b .



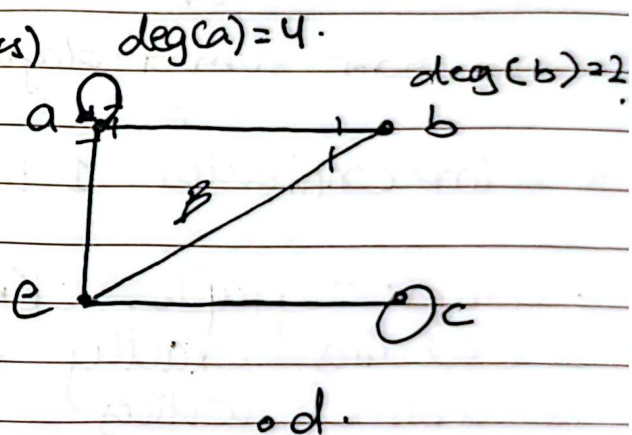
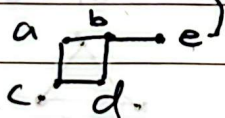
(a, b)	(a, c)
(b, a)	(b, c)
(c, b)	(c, c)

Date	Directed edges?	Multiple Edges allowed?	Loops allowed?
Simple graph	NO	NO	NO
multigraph	NO	YES	NO
pseudograph (simple)	NO	YES	YES
Directed graph	YES	NO	NO
Directed multigraph	YES	YES	NO
Mixed graph	BOTH	YES	YES

∴ For undirected Graphs:

- adjacent vertices (neighbors)
- incident (edge that connects two vertices)
- neighborhood $N(v) = \{a, b, c\}$
- $N(a) = \{v \in V \mid E(a, v) \neq \emptyset\}$
- Degree [sum of in and outs]
- Isolated [d is isolated]
- Pendant

vertex ~~can~~ connected only once to another.



... Handshake Theorem ...

Let $G = (V, E)$ be an undirected graph with e edges (e = no. of edges).
then

$$2e = \sum_{v \in V} \deg(v)$$

How many edges are there in an undirected graph with 10 vertices each of degree 6?

$$2e = 10(6)$$

$$e = 30$$

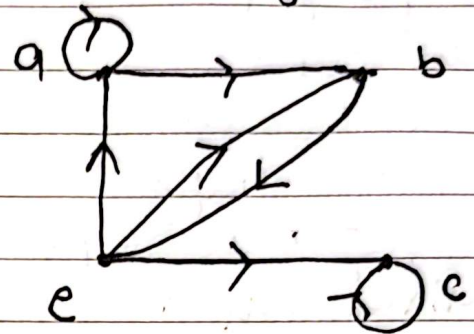
Date .

(a, b)

$$v = \deg^-(b) = 2$$

$$\deg^+(b) = 1$$

- ∴ In directed graphs:
- adjacent to - (a is adjacent to b)
 - adjacent from - (b is adjacent from a)
 - initial vertex (a)
 - terminal / end vertex (b)
 - in degree of vertex $v = \deg^-(v)$
 - out degree of vertex $v = \deg^+(v)$



$$\sum_{v \in V} \deg^-(v) = \sum_{v \in V} \deg^+(v) = |E|$$

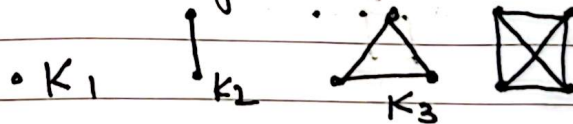
sum of in degrees
= sum of out degrees
= no. of edges.

sum - sum of degree of edges into vertex.

" " " out of "

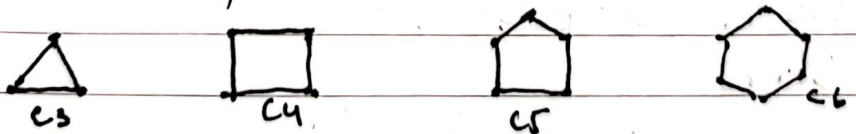
* A loop contributes 1 to each in-degree and out-degree.

- Complete graph (K_n): a simple graph that contains exactly one edges between each pair of distinct vertices.

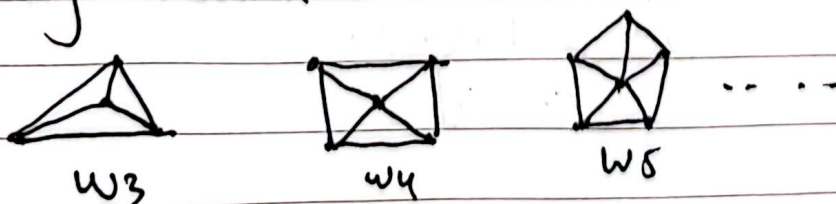


- Cycle :

A cycle (C_n), $n \geq 3$ consists of $v_1, v_2, \dots, v_n$



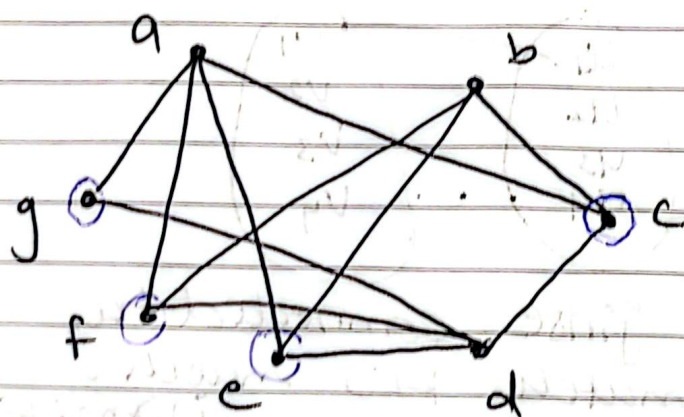
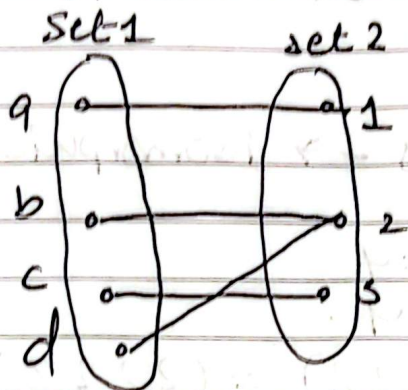
A wheel (W_n) is obtained when we add an ^{new} vertex to C_n , and connect that vertex to each existing vertex.



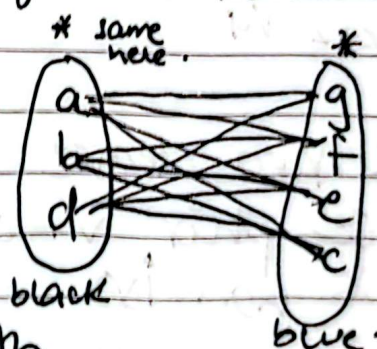
Date

Bipartite Graphs:

A simple graph is called bipartite if its vertex set, V , can be partitioned into two disjoint subsets such that each edge connects a vertex from one subset to vertex of the other.



- take a and give it a color.
- all vertices it connects to, give them different color.



set 2 vertices shouldn't be connected to each other.

- if all black vertices connect with only blue vertices and vice versa graph is bipartite. ✓

Isomorphism:

The simple graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are isomorphic if there is a one-to-one and onto function f from V_1 to V_2 with the property that a and b are adjacent in G_1 if and only if $f(a)$ and $f(b)$ are adjacent in G_2 , for all a and b in V_1 .

Steps:

1) count vertices
different? $\times$ stop.

Date

2) count edges

different? $\times$ stop.

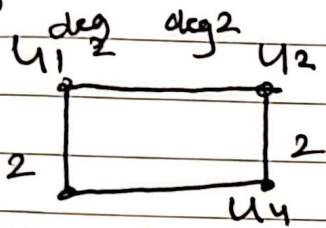
3) write degree sequence.

Different? $\times$ stop.

4) Try matching vertices.
high deg $\rightarrow$ high deg.

• preserve edges.

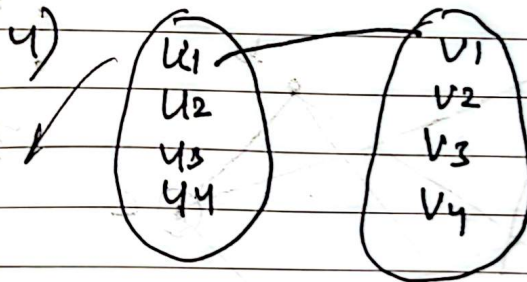
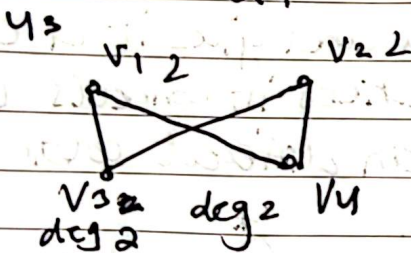
If you can match all vertices $\rightarrow$ Isomorphic $\checkmark$.



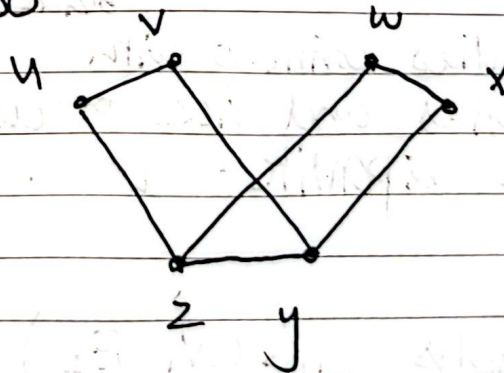
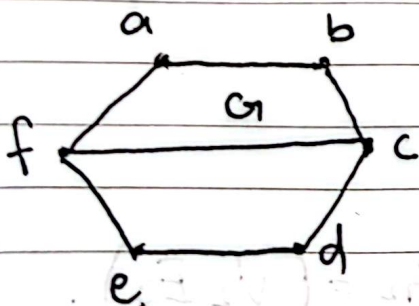
1) $4 = 4$

2) $4 = 4$

3) $(2, 2, 2, 2) = (2, 2, 2, 2)$



* u_i check kro degree aur jisse connected hai
unki degree phir v_i se match kro match $\checkmark$
match $\times$ non-iso isomorphic



vertices $= (6, 6)$

edges $= (7, 7)$